

Name: Sagi Sahasra Srihasa
Roll no: 2520030008

Step 1:

1	/* Name: Sagi Sahasra Srihasa	
2	Roll no: 2520030008	
3	*/	
4		
5	-- Step 1: Create Database	
6	CREATE DATABASE bank_management_db;	
7	USE bank_management_db;	
8	-- Step 2: Create Bank Transactions Table	
9	CREATE TABLE bank_transactions(	
10	txn_id INT PRIMARY KEY,	
11	customer_name VARCHAR(50),	
12	branch_name VARCHAR(50),	
13	transaction_type VARCHAR(20),	
14	amount DECIMAL(10,2),	
15	transaction_date DATE	
16	);	

✓	95	19:46:54	CREATE DATABASE bank_management_db	1 row(s) affected
✓	96	19:46:56	USE bank_management_db	0 row(s) affected
✓	97	19:46:59	CREATE TABLE bank_transactions(txn_id INT PRIMARY KEY, customer_name VARCHAR(50), bran...	0 row(s) affected

Step 2:

17	-- Step 3: Add Account Number Column	
18	ALTER TABLE bank_transactions	
19	ADD account_no VARCHAR(20);	
20	-- Step 4: Modify Customer Name Column	
21	ALTER TABLE bank_transactions	
22	MODIFY customer_name VARCHAR(100);	
23	-- Step 5: Rename Transaction Table	
24	RENAME TABLE bank_transactions	
25	TO customer_transactions;	

✓	98	19:47:54	ALTER TABLE bank_transactions ADD account_no VARCHAR(20)	0 row(s) affected Rec
✓	99	19:47:58	ALTER TABLE bank_transactions MODIFY customer_name VARCHAR(100)	0 row(s) affected Rec
✓	100	19:48:03	RENAME TABLE bank_transactions TO customer_transactions	0 row(s) affected

Step 3:

```
26 -- Step 6: Truncate Transaction Table
27 • TRUNCATE TABLE customer_transactions;
28 -- Step 7: Create Backup Table
29 • CREATE TABLE bank_backup(
30     txn_id INT,
31     customer_name VARCHAR(100)
32 );
33 -- Step 8: Drop Backup Table
34 • DROP TABLE bank_backup;
```

✓	101	19:48:30	TRUNCATE TABLE customer_transactions	0 row(s) affected
✓	102	19:48:35	CREATE TABLE bank_backup(txn_id INT, customer_name VARCHAR(100))	0 row(s) affected
✓	103	19:48:41	DROP TABLE bank_backup	0 row(s) affected

Step 4:

```
-- Step 9: Insert Sample Data
INSERT INTO customer_transactions
(txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
VALUES
(101, 'Harshika', 'Hyderabad', 'Deposit', 5000, '2024-01-05'),
(102, 'Pranitha', 'Hyderabad', 'Withdrawal', 2000, '2024-01-06'),
(103, 'Angela', 'Vijayawada', 'Deposit', 12000, '2024-01-08'),
(104, 'Bhanu Priya', 'Vizag', 'Deposit', 8000, '2024-01-10'),
(105, 'Nikshitha', 'Hyderabad', 'Withdrawal', 3500, '2024-01-11'),
(106, 'Manu', 'Vizag', 'Deposit', 15000, '2024-01-12'),
(107, 'Varshika', 'Vijayawada', 'Withdrawal', 1000, '2024-01-13'),
(108, 'Santhvin', 'Hyderabad', 'Deposit', 9000, '2024-01-14'),
(109, 'Ravi', 'Vizag', 'Withdrawal', 4000, '2024-01-15'),
(110, 'Lahari', 'Vijayawada', 'Deposit', 11000, '2024-01-16');
```

✓	104	19:49:25	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, trans...	10 row(s) affected Record
---	-----	----------	---	---------------------------

Step 5:

```
-- Step 10: Insert New Transaction
INSERT INTO customer_transactions
(txn_id, customer_name, branch_name, transaction_type, amount, transaction_date)
VALUES
(111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');
-- Step 11: Update Transaction Amount
UPDATE customer_transactions
SET amount = 5000
WHERE txn_id = 105;
-- Step 12: Delete Transaction
DELETE FROM customer_transactions
WHERE txn_id = 111;
```

✓	105	19:49:52	INSERT INTO customer_transactions (txn_id, customer_name, branch_name, transaction_type, amount, trans...	1 row(s) affected
✓	106	19:49:55	UPDATE customer_transactions SET amount = 5000 WHERE txn_id = 105	1 row(s) affected R
✓	107	19:49:58	DELETE FROM customer_transactions WHERE txn_id = 111	1 row(s) affected

Step 6:

```
61 -- Step 13: Display All Transactions
62 • SELECT *
63 FROM customer_transactions;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Harshika	Hyderabad	Deposit	5000.00	2024-01-05	NULL
102	Pranitha	Hyderabad	Withdrawal	2000.00	2024-01-06	NULL
103	Angela	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Bhanu Priya	Vizag	Deposit	8000.00	2024-01-10	NULL
105	Nikshitha	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL
106	Manu	Vizag	Deposit	15000.00	2024-01-12	NULL
107	Varehika	Vijayawada	Withdrawal	1000.00	2024-01-13	NULL

_transactions 1 x

Step 7;

```
64 -- Step 14: Display Deposit Transactions
65 • SELECT *
66 FROM customer_transactions
67 WHERE transaction_type = 'Deposit';
68 -- Step 15: Sort Transactions by Amount
69 • SELECT *
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
101	Harshika	Hyderabad	Deposit	5000.00	2024-01-05	NULL
103	Angela	Vijayawada	Deposit	12000.00	2024-01-08	NULL
104	Bhanu Priya	Vizag	Deposit	8000.00	2024-01-10	NULL
106	Manu	Vizag	Deposit	15000.00	2024-01-12	NULL
108	Santhvin	Hyderabad	Deposit	9000.00	2024-01-14	NULL
110	Lahari	Vijayawada	Deposit	11000.00	2024-01-16	NULL
NULL	NULL	NULL	NULL	NULL	NULL	NULL

customer_transactions 2 x

Step 8:

```
68 -- Step 15: Sort Transactions by Amount
69 • SELECT *
70 FROM customer_transactions
71 ORDER BY amount DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: |

txn_id	customer_name	branch_name	transaction_type	amount	transaction_date	account_no
106	Manu	Vizag	Deposit	15000.00	2024-01-12	NULL
103	Angela	Vijayawada	Deposit	12000.00	2024-01-08	NULL
110	Lahari	Vijayawada	Deposit	11000.00	2024-01-16	NULL
108	Santhvin	Hyderabad	Deposit	9000.00	2024-01-14	NULL
104	Bhanu Priya	Vizag	Deposit	8000.00	2024-01-10	NULL
101	Harshika	Hyderabad	Deposit	5000.00	2024-01-05	NULL
105	Nikshitha	Hyderabad	Withdrawal	5000.00	2024-01-11	NULL

Step 9:

```
72 -- Create Auditor1 User
73 • CREATE USER 'Auditor1'@'localhost'
74 IDENTIFIED BY '1234';
75
76 -- Create BranchManager User
77 • CREATE USER 'BranchManager'@'localhost'
78 IDENTIFIED BY '1234';
79 -- Step 16: Grant SELECT Permission
80 • GRANT SELECT
81 ON customer_transactions
82 TO 'Auditor1'@'localhost';
83 -- Step 17: Grant All Privileges
84 • GRANT ALL
85 ON customer_transactions
86 TO 'BranchManager'@'localhost';
87 -- Step 18: Revoke SELECT Permission
88 • REVOKE SELECT
89 ON customer_transactions
90 FROM 'Auditor1'@'localhost';
```

✓	112	19:51:49	CREATE USER 'BranchManager'@'localhost' IDENTIFIED BY '1234'	0 row(s) affected
✓	113	19:51:55	GRANT SELECT ON customer_transactions TO 'Auditor1'@'localhost'	0 row(s) affected
✓	114	19:51:58	GRANT ALL ON customer_transactions TO 'BranchManager'@'localhost'	0 row(s) affected
✓	115	19:52:02	REVOKE SELECT ON customer_transactions FROM 'Auditor1'@'localhost'	0 row(s) affected

Step 10:

```
-- Step 19: Revoke All Privileges
• REVOKE ALL PRIVILEGES
ON customer_transactions
✗ FROM 'BranchManager'@'localhost';
-- Step 20: Commit Transaction

• UPDATE customer_transactions
SET amount = 2000
WHERE txn_id = 102;
```

✓	116	19:52:31	REVOKE ALL PRIVILEGES ON customer_transactions FROM 'BranchManager'@'localhost'	0 row(s) affected
✓	117	19:52:36	UPDATE customer_transactions SET amount = 2000 WHERE txn_id = 102	0 row(s) affected Rows matched

Step 11:

```
-- Step 21: Create Savepoint
START TRANSACTION;

SAVEPOINT Before_Update;

-- Step 22: Rollback to Savepoint
UPDATE customer_transactions
SET amount = 99999
WHERE txn_id = 102;
```

✓	118	19:52:59	START TRANSACTION	0 row(s) affected
✓	119	19:53:02	SAVEPOINT Before_Update	0 row(s) affected
✓	120	19:53:06	UPDATE customer_transactions SET amount = 99999 WHERE txn_id = 102	1 row(s) affected

Step 12:

```
ROLLBACK TO SAVEPOINT Before_Update;
-- Reset Transaction State
ROLLBACK;

SET autocommit = 1;
-- Step 23: Complete TCL Example
START TRANSACTION;
```

✓	121	19:53:28	ROLLBACK TO SAVEPOINT Before_Update	0 row(s) affected
✓	122	19:53:30	ROLLBACK	0 row(s) affected
✓	123	19:53:33	SET autocommit = 1	0 row(s) affected
✓	124	19:53:35	START TRANSACTION	0 row(s) affected

Step 12;

```
117      -- Step 23: Complete TCL Example
118 •    START TRANSACTION;
119
120 •    UPDATE customer_transactions
121      SET amount = 7000
122      WHERE txn_id = 101;
123
124 •    SAVEPOINT SP1;
125
126 •    UPDATE customer_transactions
127      SET amount = 9000
128      WHERE txn_id = 102;
129
130 •    ROLLBACK TO SAVEPOINT SP1;
131
132 •    COMMIT;
```

✓	125	19:53:58	UPDATE customer_transactions SET amount = 7000 WHERE txn_id = 101	1 row(s) affected Rows matched: 1 Change
✓	126	19:54:10	SAVEPOINT SP1	0 row(s) affected
✓	127	19:54:12	UPDATE customer_transactions SET amount = 9000 WHERE txn_id = 102	1 row(s) affected Rows matched: 1 Change
✓	128	19:54:14	ROLLBACK TO SAVEPOINT SP1	0 row(s) affected
✓	129	19:54:16	COMMIT	0 row(s) affected